INCH-POUND

MIL-C-80090D
6 February 1992
SUPERSEDING
MIL-C-80090C
17 June 1985

MILITARY SPECIFICATION

CUTTING SYSTEMS, METAL, PLASMA ARC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers cutting systems which utilize plasma arc as the cutting medium. The systems shall be furnished with the equipment and accessories specified herein.

1.2 Classification. This specification covers cutting systems of the following sizes. The size to be furnished shall be as specified (see 6.2 and table I).

Size 14	-	14 Amperes DC,	20 percent duty cycle.
Size 20	-	20 Amperes DC,	20 percent duty cycle.
Size 30	-	30 Amperes DC,	40 percent duty cycle.
Size 40	-	40 Amperes DC,	50 percent duty cycle.
Size 50	-	50 Amperes DC,	60 percent duty cycle.
Size 60	-	60 Amperes DC,	60 percent duty cycle.
Size 75	-	75 Amperes DC,	70 percent duty cycle.
Size 80	-	80 Amperes DC,	80 percent duty cycle.
Size 100	-	100 Amperes DC,	80 percent duty cycle.
Size 150	-	150 Amperes DC,	50 percent duty cycle.
Size 200	-	200 Amperes DC,	100 percent duty cycle.
Size 250	-	250 Amperes DC,	100 percent duty cycle.
Size 400	-	400 Amperes DC,	100 percent duty cycle.
Size 500	-	500 Amperes DC,	100 percent duty cycle.
Size 1000	-	1,000 Amperes DC,	100 percent duty cycle.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, 2163 Airways Blvd. Memphis, Tennessee 38114-5213, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3433

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:37Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-W-45562 - Welding and Soldering Equipment, Supplies and Accessories, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

(Unless otherwise indicated, copies of Federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U.S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001).

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contract activity).

Source: https://assist.dla.mil -- Downloaded: 202610-23T04:37Z
Check the source to verify that this is the current version before use.

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code (DOD Adopted).

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101).

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services).

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), one complete system shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.2 Design. The systems shall be designed for cutting metals by using an electric arc and air or mixtures of appropriate gases such as argon, hydrogen, or nitrogen to form a plasma cutting arc. Water injection into or in the vicinity of the torch nozzle may be used to improve the arc and the quality of cut. The systems shall be new (not a prototype) and one of the manufacturer's current production models capable of operations in accordance with the requirements herein. The systems shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, or repair.

3.2.1 Measuring and indicating device calibration. Measuring and indicating devices shall be graduated in inch-pound (see 6.5.1), metric (see 6.5.2), or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.2 Reclaimed materials. The systems may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the systems. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, rebuilt, or remanufactured end products are allowed under this specification.

Source: https://assist.dla.mil -- Downloaded: 202610-23T04:37Z
Check the source to verify that this is the current version before use.

3.2.3 Energy efficiency. The systems and components that directly consume energy in normal operation shall be designed and constructed for energy efficiency in accordance with the developments available within the industry.

3.2.4 Safety and health requirements. All safety devices normally incorporated in this equipment by the manufacturer for the protection of the operator shall be furnished. The outer shell of the plasma torch assembly shall be electrically insulated to ensure that the operator will not be exposed to electrically charged components. A safety device shall be provided to disconnect electrical power from the torch in the event the cooling medium or plasma gas falls below a safe operation level. All parts, components, mechanisms, and assemblies furnished on the systems, whether or not specifically required herein, shall conform with all requirements of 29 CFR 1910. Additional safety and health requirements shall be as specified in detail (see 6.2).

3.2.5 Environmental protection. During the manufacture, operation, service, transportation, or storage of the systems and components, the systems shall meet all applicable Environmental Protection Agency (EPA) and OSHA restrictions for materials classified as hazardous to the environment in effect on the date of the contract. Specific environmental protection requirements exceeding the established EPA and OSHA regulations shall be as specified by the procuring activity (see 6.2).

3.2.6 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. All parts shall be manufactured to such standards as will permit replacement or adjustment without modification of parts or the systems components.

3.3 Construction. The systems shall be constructed of parts which are new, without defects, and free of repairs. The systems shall withstand all forces encountered during operation at the maximum rating and capacity without permanent distortion or failure.

3.3.1 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.2 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.3 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.4 Painting. Unless otherwise specified (see 6.2), the systems shall be painted in accordance with the manufacturer's commercial practice and color.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:37Z
Check the source to verify that this is the current version before use.

3.3.5 Threads. All threaded parts used on the systems and related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.6 Electromagnetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystem(s) class and the emission and susceptibility requirements shall be as specified.

3.4 Components.

3.4.1 Power supply. Unless otherwise specified, the power supply furnished may be a transformer-rectifier type, inverter type, or any type designed for plasma arc cutting purposes. When a specific type is required, the type furnished shall be as specified by the procuring activity (see 3.8 and 6.2). The rated direct current (DC) output and duty cycle shall be not less than stated in paragraph 1.2 for the size specified. Unless otherwise specified (see 6.2), sizes 14 through 50 systems may be furnished with or without provisions for varying the output current. Sizes 60 thru 1,000 systems shall have provisions for controlling the output current from not greater than 15 volts or 25 percent of full output, whichever is greater, to 100 percent output. The output shall be capable of being controlled to an accuracy of $\pm$ 5 percent. This may be accomplished by the use of a panel mounted ammeter or calibrated output control. The power supply shall be enclosed in a cabinet which has the physical strength to provide support and protection of all components and shall be designed to provide access for servicing. The power supply shall be capable of operating under full load in an ambient temperature up to 115 degrees Fahrenheit (F). When two or more power supplies are furnished for parallel operation, all necessary connecting cables and hardware shall be furnished. When specified (see 6.2), the power supply shall be furnished mounted on wheels.

3.4.1.1 Pilot arc. The cutting systems shall be equipped with a device for generating a pilot arc.

3.4.2 Torches. The torches, when furnished, shall be gas or liquid cooled. Replacement, positioning, and adjustment of the electrode, nozzle, and associated parts shall be accomplished without the use of special tools, except as supplied with the system.

3.4.2.1 Machine torch. When specified (see 6.2), a machine torch designed for mounting on a carriage mechanism capable of plasma arc cutting speeds shall be furnished. The torch shall be designed for operation with the DC power supply specified in 3.4.1. The type and quantity of nozzles to be furnished and the type and thickness of metal to be cut shall be as specified (see 6.2).

3.4.2.2 Manual torch. When specified (see 6.2), a manual type (hand held) torch shall be furnished. The torch shall be designed for operation with the DC power supply specified in 3.4.1. A switch shall be located on the handle to control operation of the cutting system. Unless otherwise specified (see 6.2), the torch shall have a 70 or 75 degree head angle. The type and quantity of nozzles to be furnished shall be as specified (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 202610-23T04:37Z
Check the source to verify that this is the current version before use.

3.4.3 Torch leads and hoses. All DC power leads, gas and water hoses, and all connectors or fittings necessary for operation of the cutting systems at full rated power shall be furnished. Unless otherwise specified (see 6.2), the torch leads and hoses which connect directly to the torch shall be not less than 25 feet in length. All other interconnecting leads and hoses shall be the manufacturers standard length.

3.4.4 Controls. All controls, gauges, and indicating lights necessary for complete operation of the cutting system shall be furnished.

3.4.5 Liquid cooling system. When specified (see 6.2), a recirculating type cooling system with the capability to provide cooling for 100 percent duty cycle operation of the torch at full rated power shall be supplied. When specified (see 6.2), the system shall be refrigerated.

3.4.6 Water muffler. When specified (see 6.2), a water muffler shall be furnished. The muffler shall be complete and ready for operation.

3.4.7 Electrical system. The electrical system shall conform to NFPA 70. Unless otherwise specified (see 6.2), sizes 14 and 20 cutting systems shall operate from 120 volt, single phase; sizes 30, 40, and 50 shall operate from 220, single phase; and sizes 60 through 1000 shall operate from 230/460 volts 3 phase. Sizes 60 thru 1000 shall be initially wired as specified (see 6.2). All cutting systems shall operate from a 60 Hz source.

3.4.7.1 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type.

3.4.7.2 Plasma and cooling gases. If specific plasma and cooling gases are required, they shall be as specified (see 6.2).

3.5 Size and capacity. Unless otherwise specified (see 6.2), the size and capacity of the cutting systems shall meet the requirements of 1.2 and table I for the size specified. The size and capacity shall not be less than the stated requirements for the size specified. Unless otherwise specified (see 6.2), a larger size may be offered.

3.6 Performance. The cutting systems shall meet all requirements specified herein. Cutting speeds of the systems shall be not less than specified in table I. The cut shall be dross free or if dross is present, easily removed.

3.6.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2).

3.7 Hour meter. When specified (see 6.2), the systems shall be equipped with an hour meter. The hour meter shall be installed to display accumulated arc time. The meter shall be of the nonresetting type and shall have a range of not less than 0 to 9,999 hours in increments of not greater than 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the systems.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:37Z
Check the source to verify that this is the current version before use.

TABLE I. Cutting speed (inches per minute).

Size	Material: Low carbon steel (1020 - 1030)							
	12 gage	1/8 in	1/4 in	3/8 in	1/2 in	3/4 in	1 in	2 in
14	25	--	--	--	--	--	--	--
20	30	--	--	--	--	--	--	--
30	--	35	--	--	--	--	--	--
40	--	--	30	--	--	--	--	--
50	--	--	35	--	--	--	--	--
60	--	--	--	25	--	--	--	--
75	--	--	--	40	25	--	--	--
100	--	--	--	--	25	--	--	--
150	--	--	--	--	30	--	--	--
250	--	--	--	--	--	28	--	--
400	--	--	--	--	--	40	--	--
500	--	--	--	--	--	--	50	--
1,000	--	--	--	--	--	--	--	25

3.8 Physical size limitations. When specified (see 6.2), the physical size of the systems shall be limited. The physical size of the systems shall not exceed the height, width, and length restrictions specified by the procuring activity.

3.9 Standard equipment. All standard equipment normally provided with the manufacturer's commercial system shall be furnished.

3.10 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2). The optional equipment provided shall be functional without requiring modification of the equipment or systems.

3.11 Repair parts. Required repair parts shall be furnished as specified (see 6.2).

3.12 Marking on instruments, control panels, charts, and plates. All words on the systems including all instruments, control panels, control units, charts, plates, and related attachments shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:37Z
Check the source to verify that this is the current version before use.

3.12.1 Nameplate. A nameplate shall be securely attached to each system. Unless otherwise specified (see 6.2), the nameplate shall contain the information listed below.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract number or order number
National stock number or plant equipment code
Date of manufacture

3.13 Workmanship. Workmanship of the system and its accessories shall meet all requirements specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

c. Acceptance test (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.7.1.

Source: https://assist.dla.mil -- Downloaded: 202610-23T04:37Z
Check the source to verify that this is the current version before use.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.6, and all tests in 4.7. The cutting system shall pass the examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each system prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.6, the test in 4.7.2, and the inspection in 4.8. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Examination. Each system shall be examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 41° to 104° Fahrenheit (F) and 20 to 95 percent relative humidity.

4.7.2 Operational test. The cutting system shall be connected to the specified input power, primary gas, and cooling medium. All components of the system shall be operated as required to assure proper operation of the complete cutting system. The starting arc shall be tested for proper operation. All meters, gauges, and pilot lights shall be observed for proper operation. When applicable, the water circulating system shall be checked to assure proper operation.

4.7.3 Performance tests. The complete system utilizing a cutting gas and a torch selected by the manufacturer shall be tested to determine conformance with 3.5. This shall be accomplished by cutting a 24 inch long piece of low carbon steel of the thickness specified in table I for the size cutting system being tested. The cutting speed of the system shall meet or exceed the cutting requirements of table I.

4.7.4 Additional performance tests. Performance tests required in addition to or in lieu of the requirements of 4.7.3 shall be as specified and fully described (see 6.2).

4.7.5 Electromagnetic interference control test. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8 using the procedures given in MIL-STD-462.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-W-45562.

Source: https://assist.dla.mil -- Downloaded: 202640-23T04:37Z
Check the source to verify that this is the currentversion before use.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended use. The plasma arc cutting systems described in this specification will be used for cutting metal, using a hand held torch or a machine mounted torch.

6.2 Ordering data.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Size required (see 1.2).

c. Issue of DODISS to be cited in the solicitation and, if required, the specification issue of the individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. System of units for measuring and indicating devices (see 3.2.1).

f. Additional safety and health requirements, if required (see 3.2.4).

g. Environmental protection requirements exceeding EPA and OSHA requirements (see 3.2.5).

h. Painting, if different (see 3.3.4).

i. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.6).

j. Specific type power supply, if required (see 3.4.1).

k. Output current control for sizes 14 thru 50, if required (see 3.4.1).

l. Power supply mounted on wheels, if required (see 3.4.1).

m. Machine torch, if required specify type and quantity of nozzles required, type and thickness of material to be cut, (see 3.4.2.1).

n. Manual torch, if required specify type and quantity of nozzles required and type and thickness of material to be cut (see 3.4.2.2).

Source: https://assist.dla.mil -- Downloaded: 202610-23T04:37Z
Check the source to verify that this is the current version before use.

o. Angle of torch head, if different (see 3.4.2.2).

p. Length of power cables and hose assemblies, if different (see 3.4.3).

q. Recirculating cooling system, if required (see 3.4.5).

r. Refrigerated cooling system, if required (see 3.4.5)

s. Water muffler, if required (see 3.4.6).

t. Voltage requirements, if different (see 3.4.7).

u. Initial voltage for sizes 60 thru 1,000, as specified (see 3.4.7).

v. Plasma and cooling gas requirements (see 3.4.7.2).

w. Size and capacity, if different. (see 3.5).

x. If specified, larger size shall not be accepted. (see 3.5).

y. Additional performance requirements, if required (see 3.6.1).

z. Hour meter, if required (see 3.7).

aa. Specify physical size limitation, if required (see 3.8).

bb. Specify and fully describe optional equipment, if required (see 3.10).

cc. Specify applicable repair parts, if required (see 3.11).

dd. Nameplate, if different (see 3.12.1).

ee. Inspection conditions if different (see 4.3).

ff. First article inspection, if different (see 4.4).

gg. Quality conformance inspection, if different (see 4.5).

hh. Additional performance tests, if required (see 4.7.4).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specification acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:37Z
Check the source to verify that this is the current version before use.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
N/A	DI-TMSS-80527	Commercial Off-The-Shelf (COTS) Manuals	Require all data be in the English Language
N/A	DI-TMSS-80528	Supplemental Data for Commercial Off-The-Shelf (COTS) Manuals	Require all data be in the English Language

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.3.1 Technical data. It should be specified in the remarks section of the DD Form 1423 that all technical data furnished should be written in the English language.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Institute of Standards and Technology. Inch-pound units having the same names in other countries may differ in magnitude.

6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le Systeme International d'Unites (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

6.6 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:37Z
Check the source to verify that this is the current version before use.

6.7 Cross-reference of classification. Classification changes from the previous issue of this specification are as follows:

MIL-C-80090C	MIL-C-80090D
	Size 14(new size)
	Size 20(new size)
Size 1	Size 30(new size)
	Size 40(new size)
Size 2	Size 50
	Size 60(new size)
	Size 75(new size)
	Size 80(new size)
	Size 100(new size)
Size 3(deleted)	
	Size 150(new size)
	Size 200(new size)
Size 4	Size 250
Size 5	Size 400
Size 6	Size 500
Size 7	Size 1,000

6.8 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.9 Safety and health requirements. OSHA 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

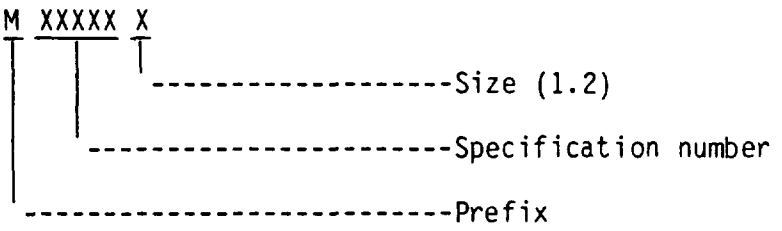
6.10 Training. Training required by the procuring activity should be provided as specified in the contract.

6.11 Warranty. Warranty requirements should be as specified by the procuring activity in the contract.

6.12 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:37Z
Check the source to verify that this is the current version before use.

6.13 Part or Identifying Numbers (PINs). The PINs to be used for systems acquired to this specification are created as follows:



6.14 Subject term Key word listing.

Arc cutting
Gas cutting
Metal cutting
Torches

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Army - GS
Navy - MC
Air Force - 84

User activities:
Army - AV, AR

Preparing Activity:
DLA - IP

Project (3433-0122)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:37Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-80090C	92/02/06

3. DOCUMENT TITLE
CUTTING SYSTEMS, METAL, PLASMA ARC

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
c. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial (2) AUTOVON (*If applicable*)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (*Include Area Code*)	
Defense Industrial Plant Equipment Center ATTN: DIPEC-SSG /Mr. Threadgill	(1) Commercial (901)-775- 4749	(2) AUTOVON 683- 4749
c. ADDRESS (*Include Zip Code*) 2163 Airways Blvd. Memphis, TN 38114- 5213	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:37Z
Check the source to verify that this is the current version before use.